M9.3.b.1 Semantic Web and Ontologies

Tuto/Lab 2: OWL SWRL Reasoning

Part 1: Work environment

1- Download and install the last version of Protégé tool at
https://protege.stanford.edu
Ensure SWRL Tab plugin is there.

Part 2: Importing and inspecting a small ontology

Before starting you may have a look on the following demo
https://www.youtube.com/watch?v=I81pLmK_lHg&list=PLNXdQl4kBgzubTOfY5cbtxZCgg9UTe-uF&index=58

1- Replay the demo steps on Protégé. The considered Pizza ontologie is provided on Celene PizzaSWRL.ttl.

2- Similarly, consider the Viggi ontology introduced in the course exercise and available on Celene Viggi.ttl, add the rules, and perform the reasoning on Protégé.

Part 3: Ontology enrichement and reasoning

In this part you should consider university.ttl ontology from the previous lab and use SWRL rules to solve questions 4 to 9 in an easier way than using OWL axioms. You can alternatively retake the provided ontology and add classes and properties of questions 1 to 3, or create a copy of your last ontology version and remove the parts corresponding to questions 4 to 9.

For the following sentences, write the SWRL rule, run Drools reasoner, and check the obtained results. Do not hesitate to add new classes, properties, and individuals to check the effectiveness of some rules.

1- An ExpertTeacher is a Teacher having one or more expertise topics.
2- An ExpertTeacher for a specific Course is the author of a book of the same topic as the course.
3- A teacher who supervises StudentA also supervises any student working with StudentA.
4- A teacher A helpfullyAdvices a student when he advice this student who in turn is supervised by teacher B, collegue of teacher A.
5- A course has the same topics as its referenced materials
6- Discuss rules chaining for the case of ExpertTeacher (rule 2 and rule 5).
7- In many cases, two authors who wrote the same book are considered co-authors.
8- A teacher with more than one topic of expertise may be inferred as a SeniorExpertTeacher.
9- A student is “SuccessfulStudent” if their grade is greater than or equal to 15.

10- A course containing “AI” in its label is tagged as an AICourse.

Part 4: Further reading

1- Read the two scientific papers related to rule-based reasoning in dynamic knowledge bases.
   a. Incremental Update of Datalog Materialisation: The Backward/Forward Algorithm, Motik et al. 2014
   b. Web Reasoning Using Fact Tagging, Terdjimi et al. 2018
   c. Incremental Reasoning for Dynamic ABox Materialization Based on Subgraph, Zhu et al. 2022